Evidence and attribution of the land carbon sink's historic enhancement

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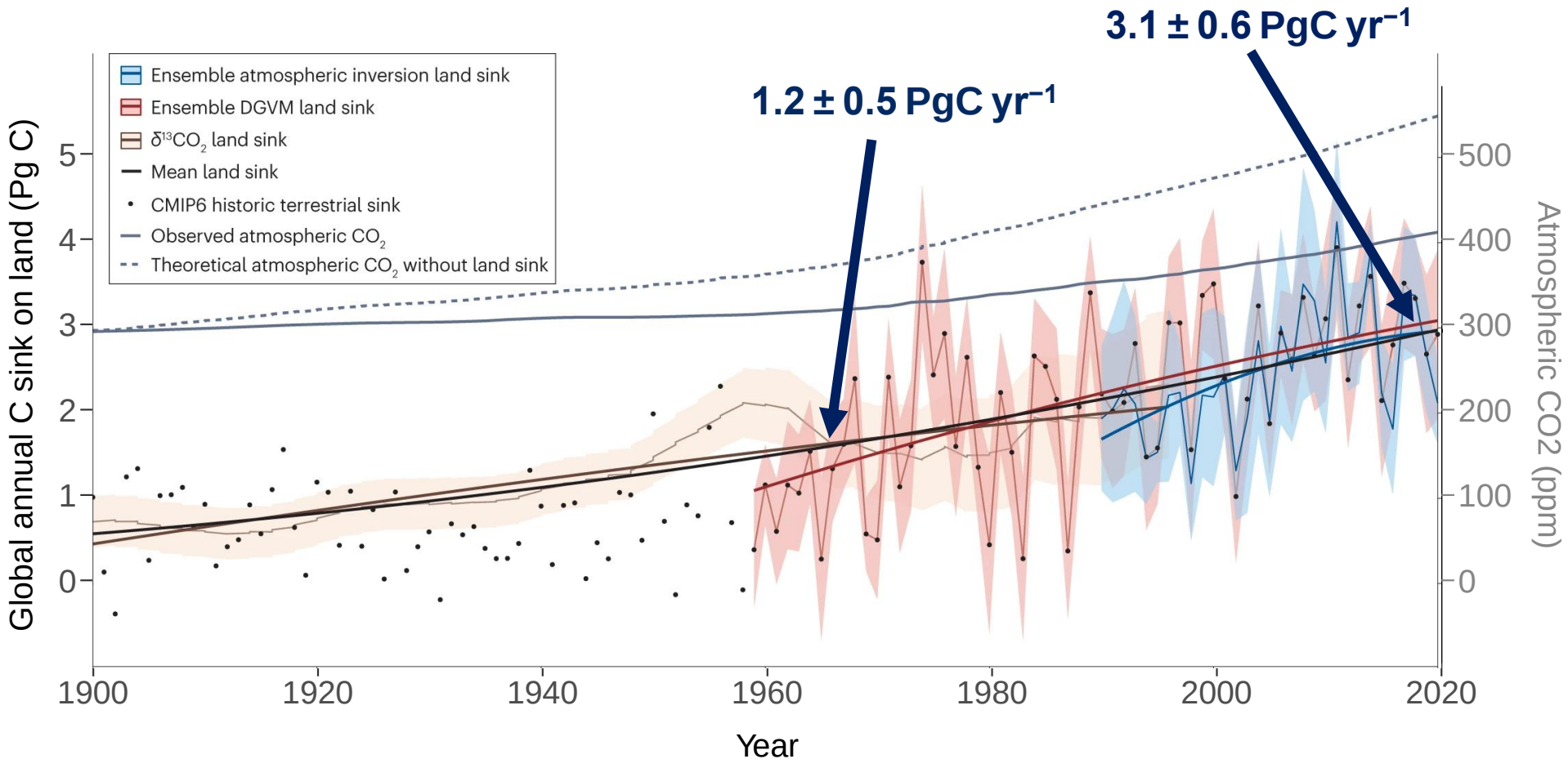
Multiple estimates indicate that the **global annual land carbon sink has increased**



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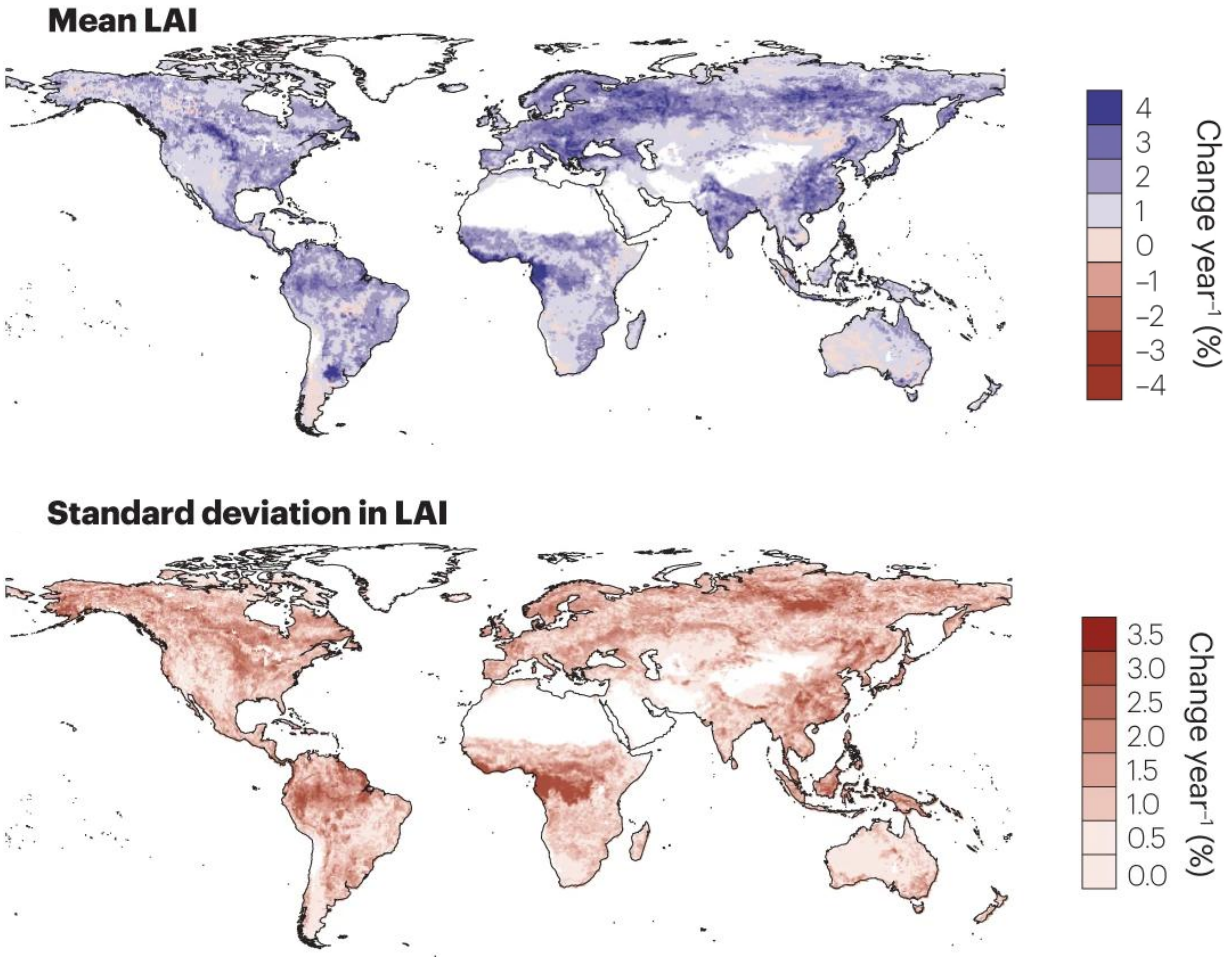
Remote sensing products suggest **widespread increases in leaf area index**



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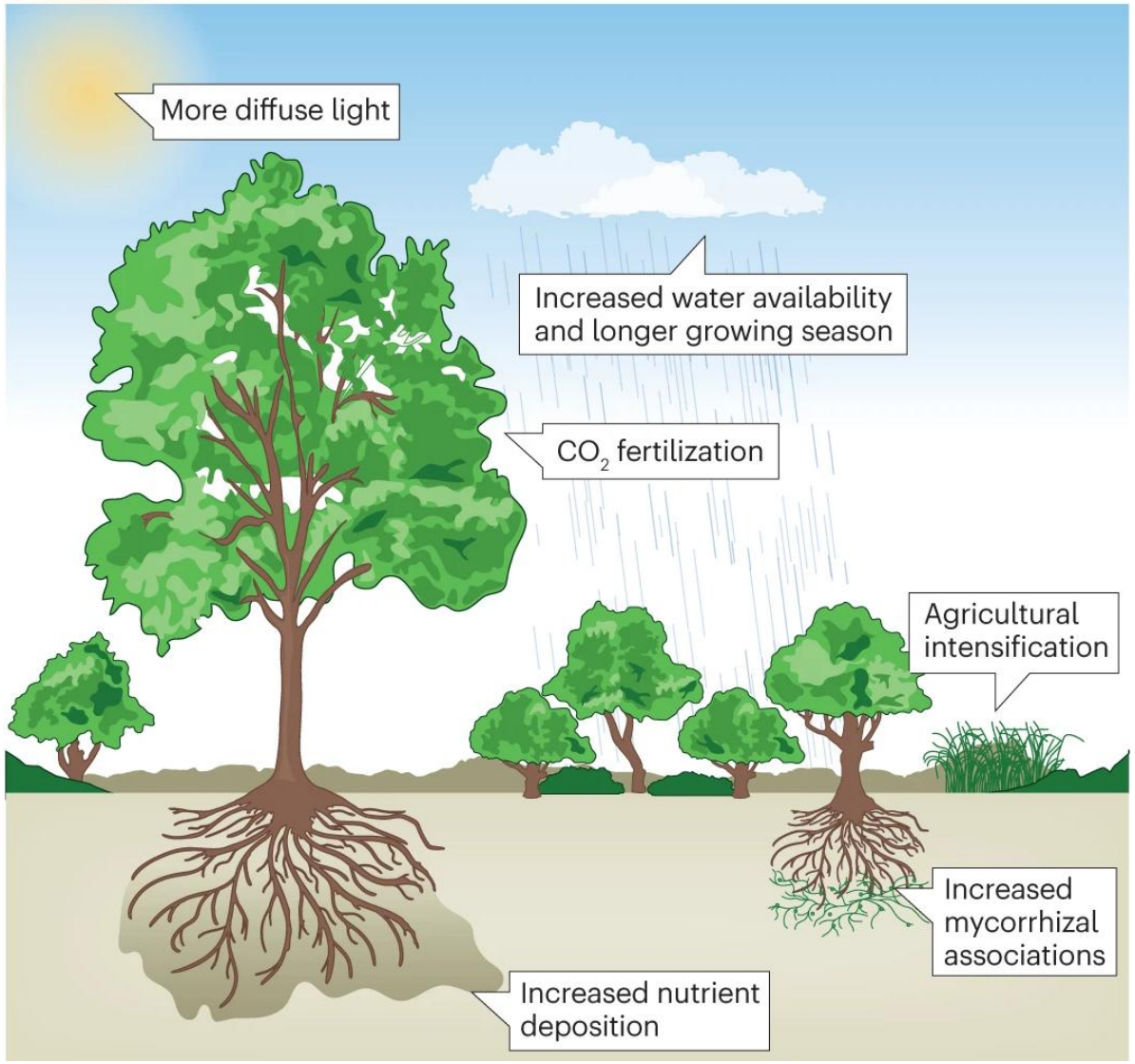
The land carbon sink is **enhanced via multiple pathways**…



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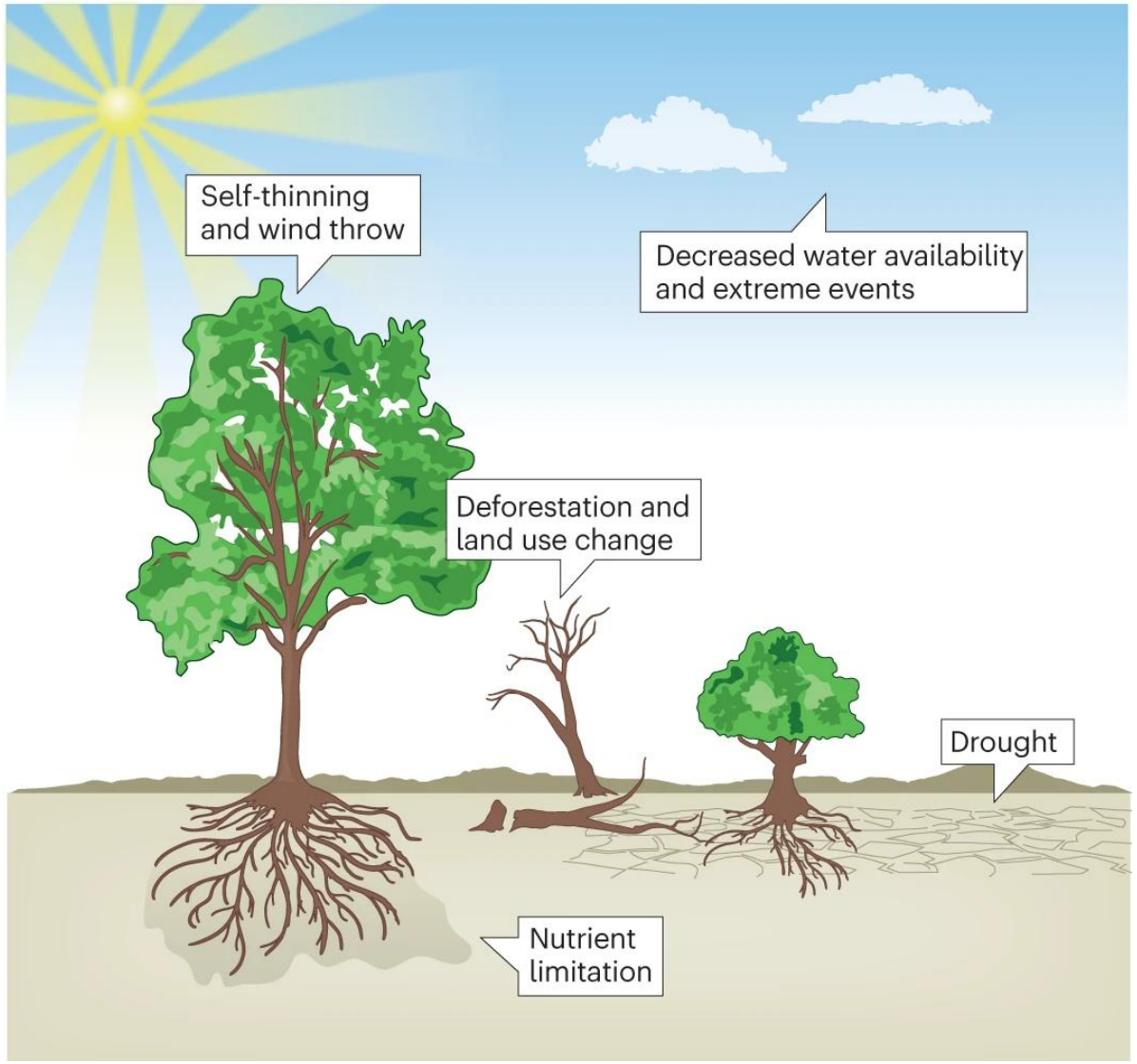
…and **reduced via concurrent dynamics**



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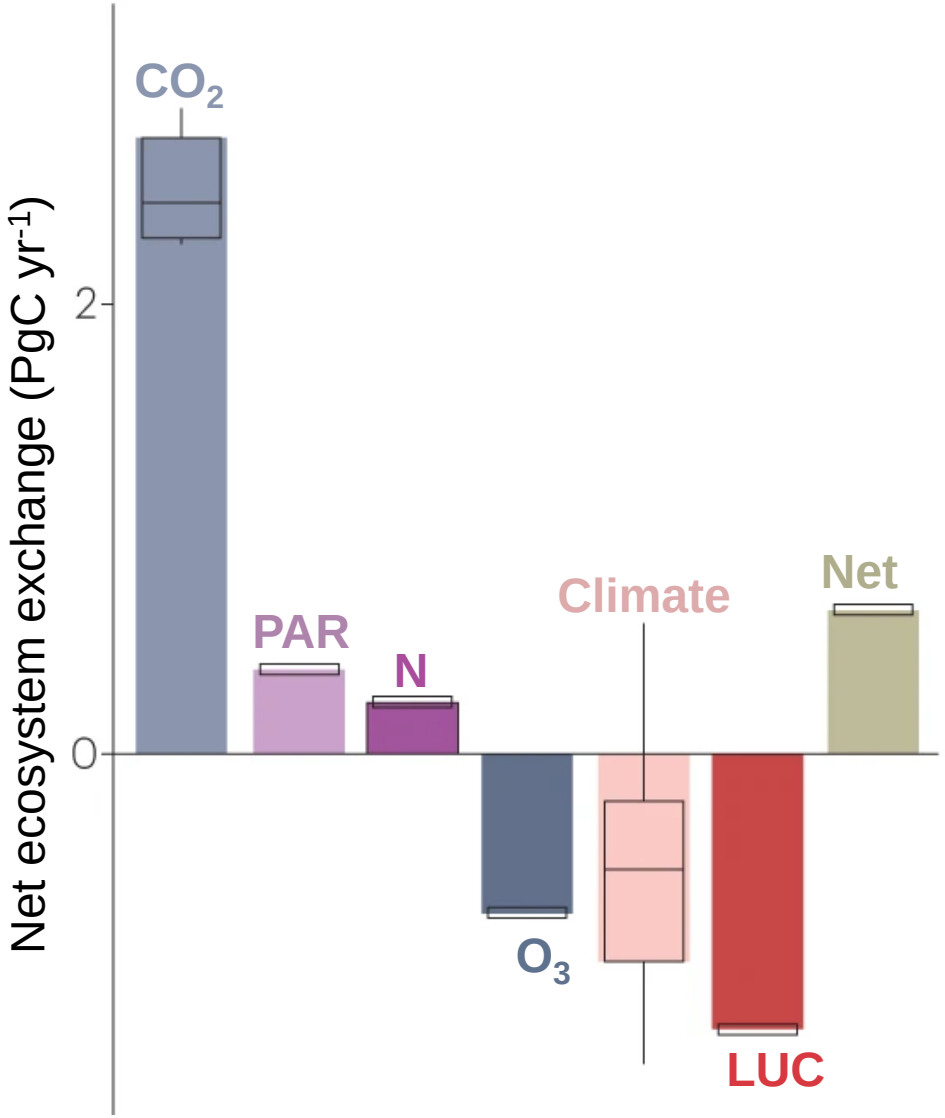
CO_2 fertilization resulted in net carbon sink enhancement, despite decreases from land use change and climate



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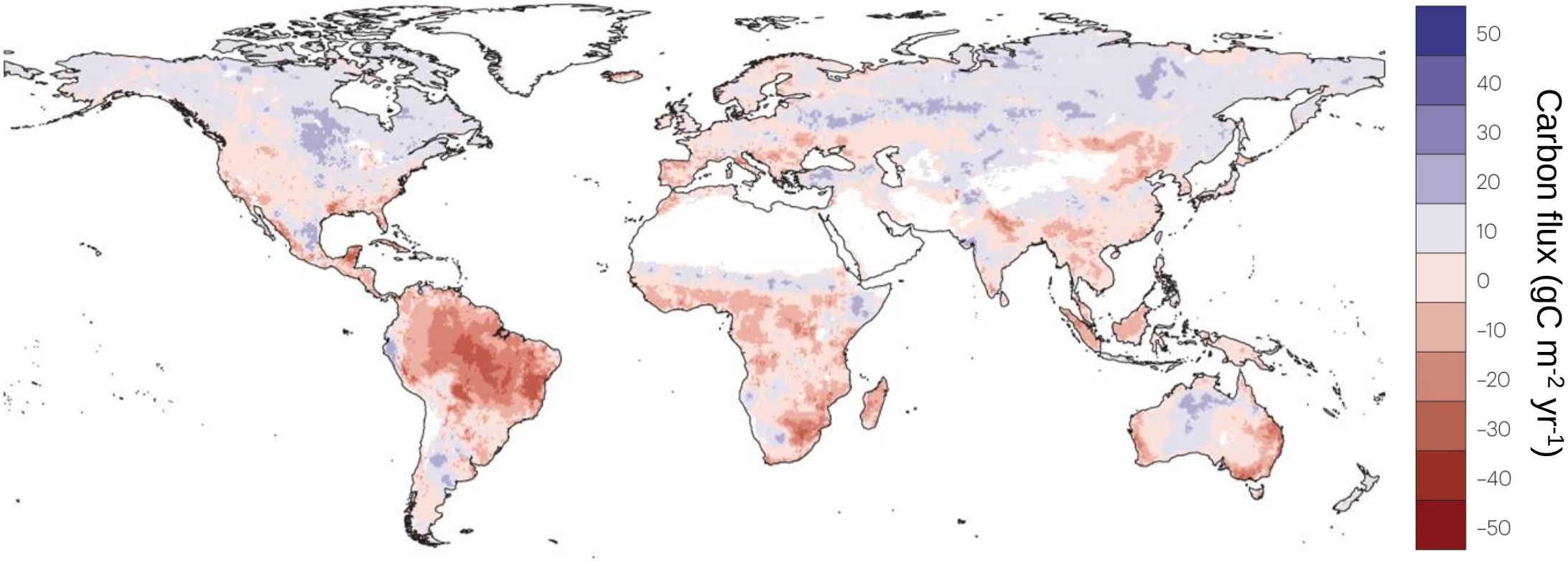
Climate **increased the sink at high latitudes** and **decreased the sink in the tropics**



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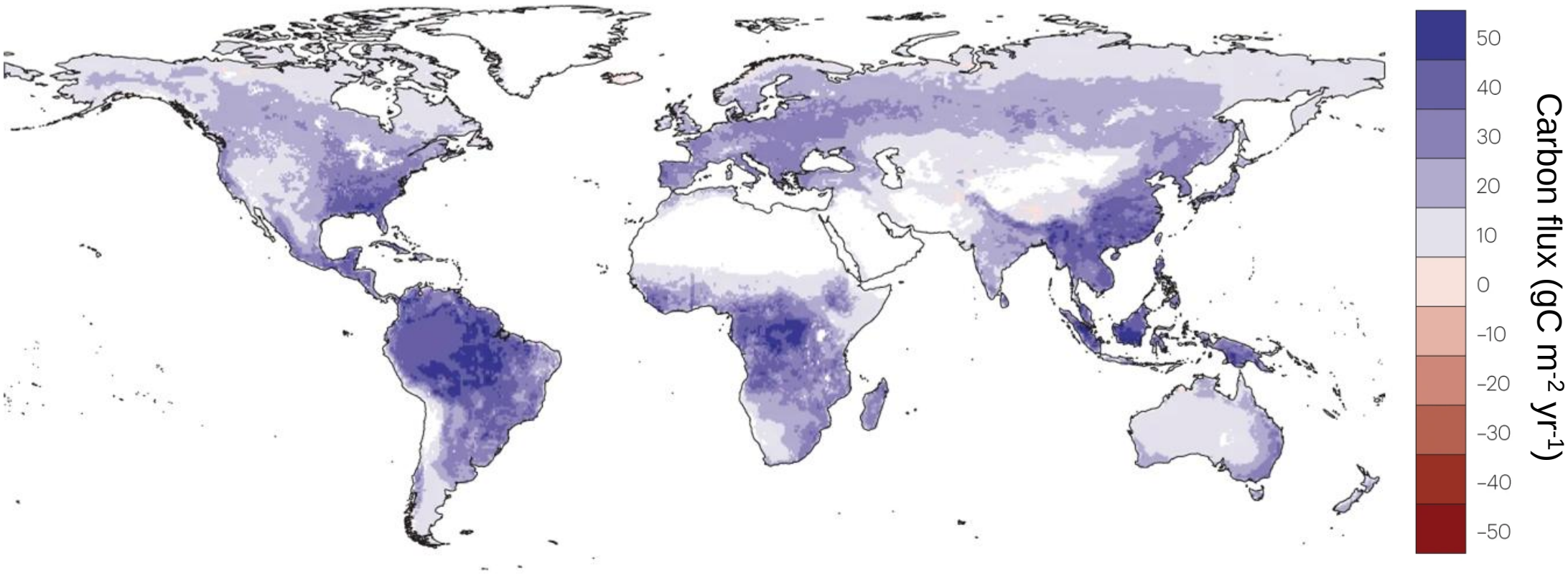
CO_2 fertilization **increased the global carbon sink**



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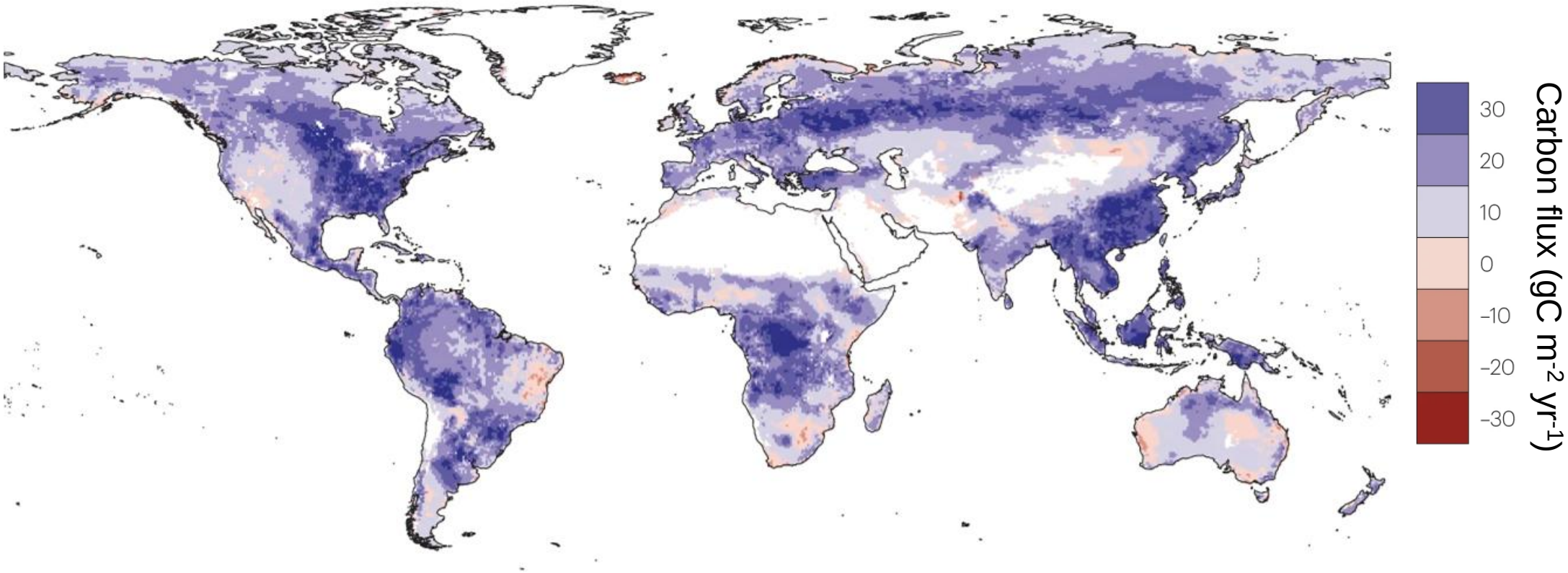
Together, **CO_2 and climate enhanced the land carbon sink globally**, with decreases in some regions



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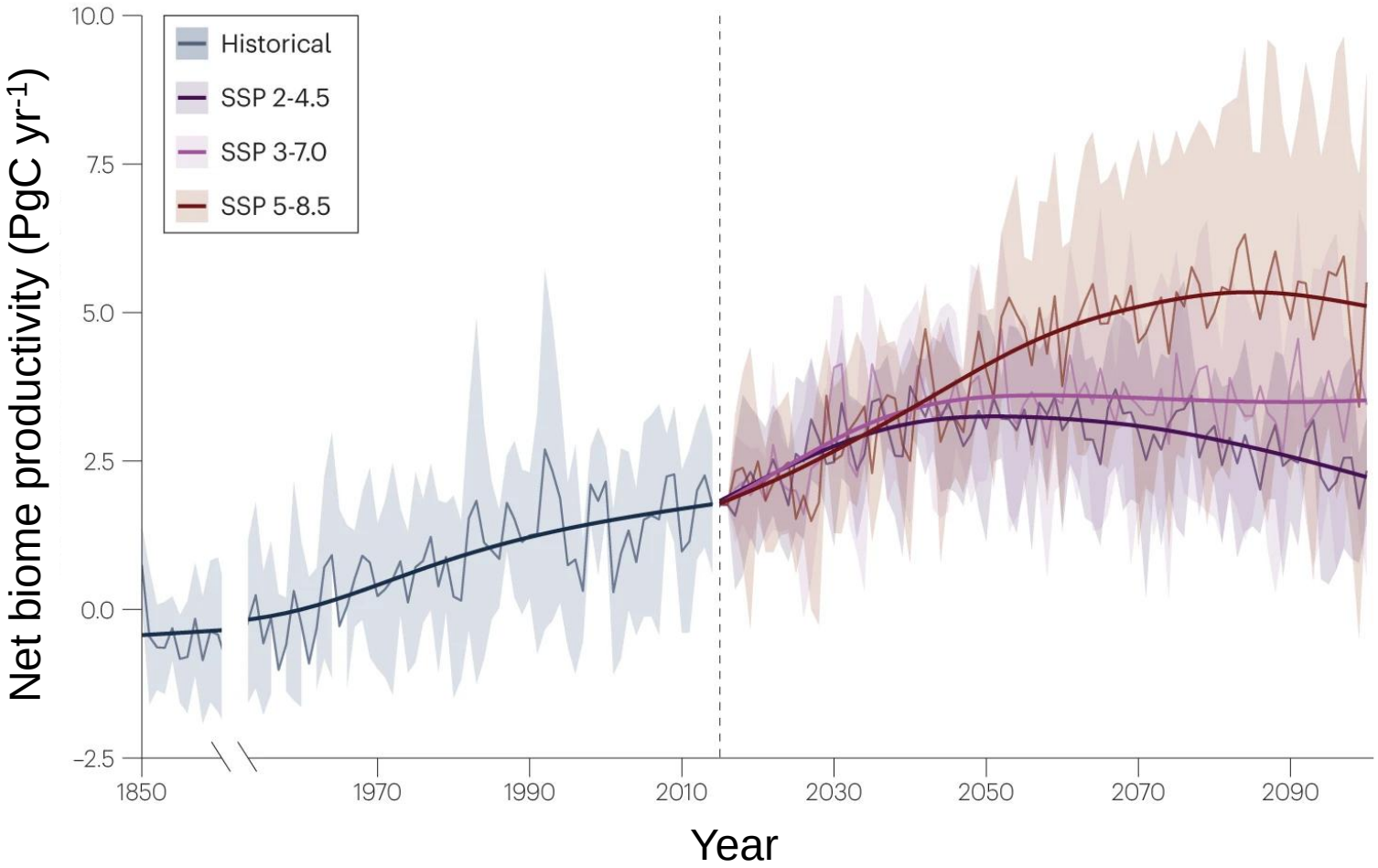
CMIP6 projections suggest **land carbon sink enhancement with saturate by 2100**



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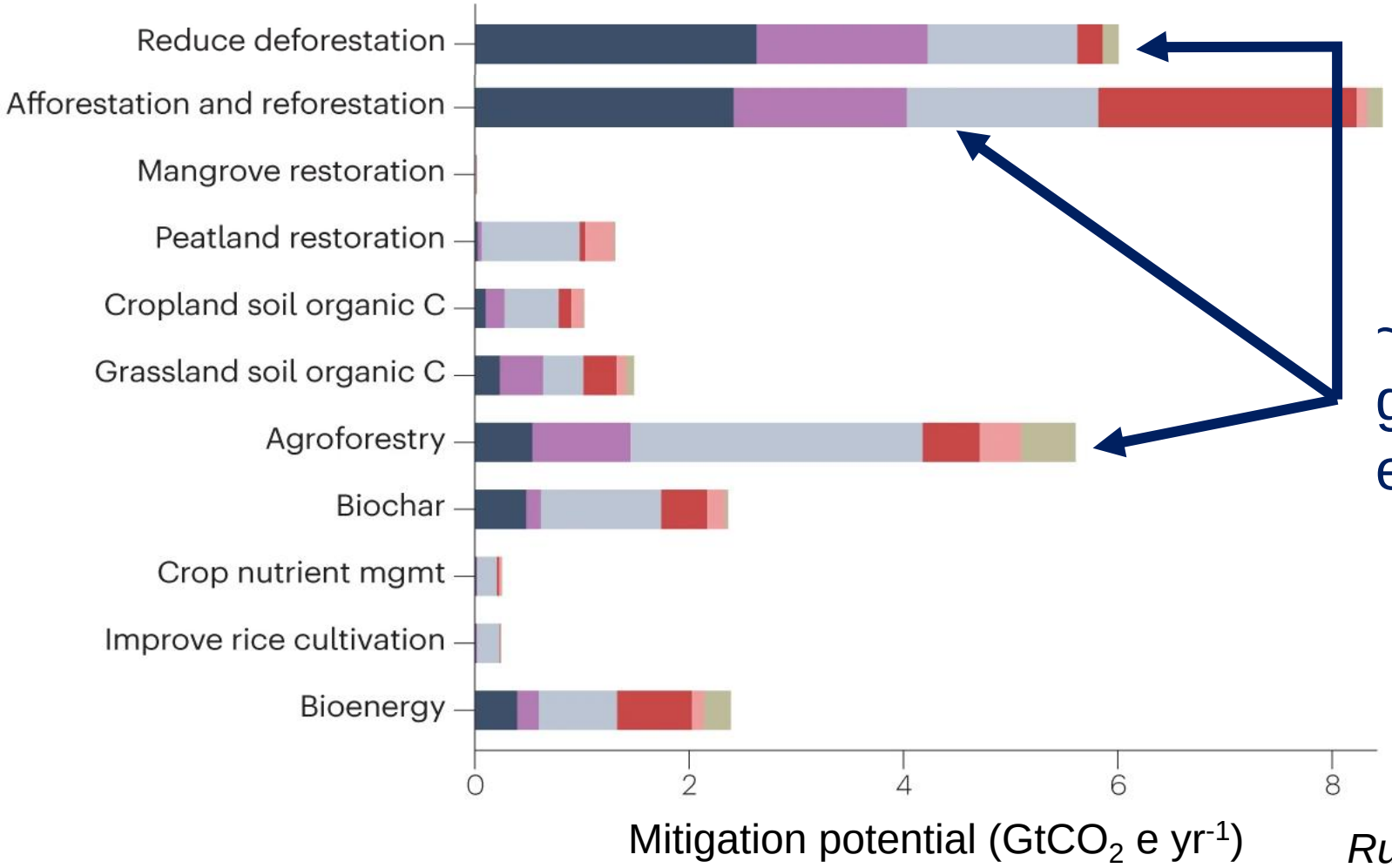
Land management strategies have **large potential to mitigate climate change**…



~50% of annual global CO_2 emissions

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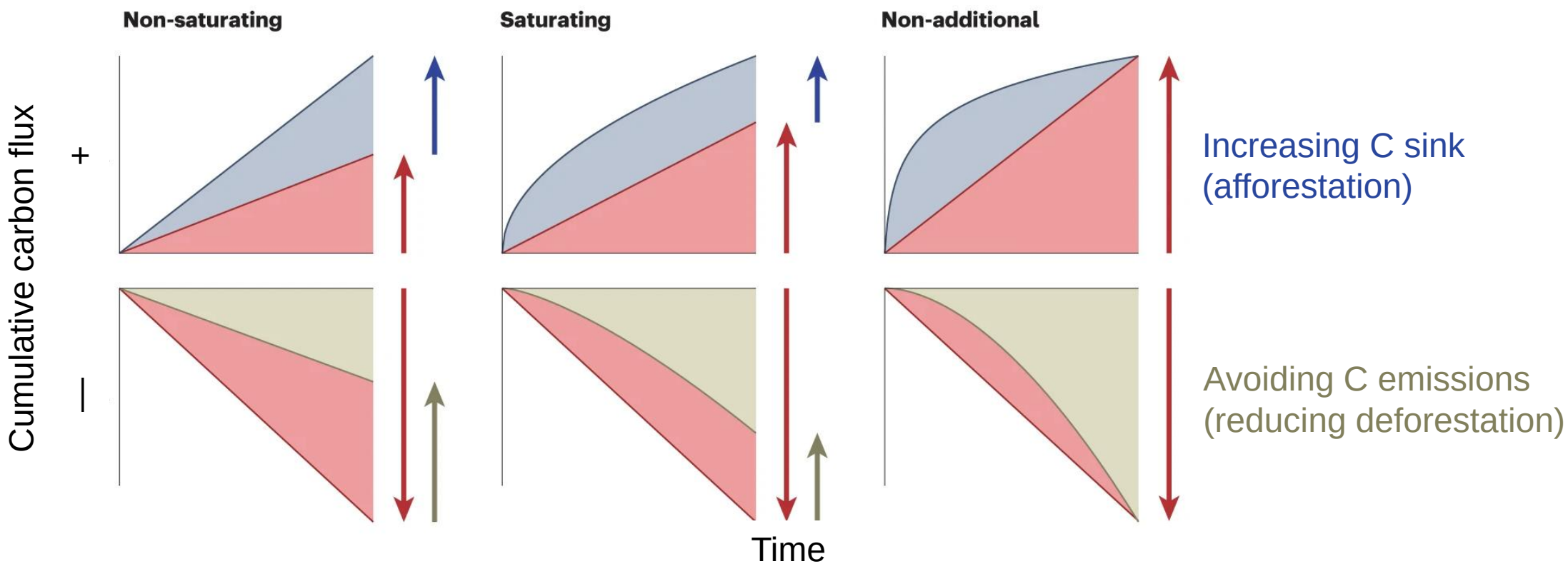
Adapted from Roe et al. GCB, 2021

…but longevity of **nature-based climate mitigation strategies are highly uncertain**



Ruehr et al. Nature REE, 2023
Adapted from Novick et al., Indiana University, 2022

RESEARCH ARTICLE | SOCIAL SCIENCES |

Overstated carbon emission reductions from voluntary REDD+ projects in the Brazilian Amazon

Thales A. P. West, Jan Börner, Erin O. Sills, and Andreas Kontoleon Authors

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The Guardian

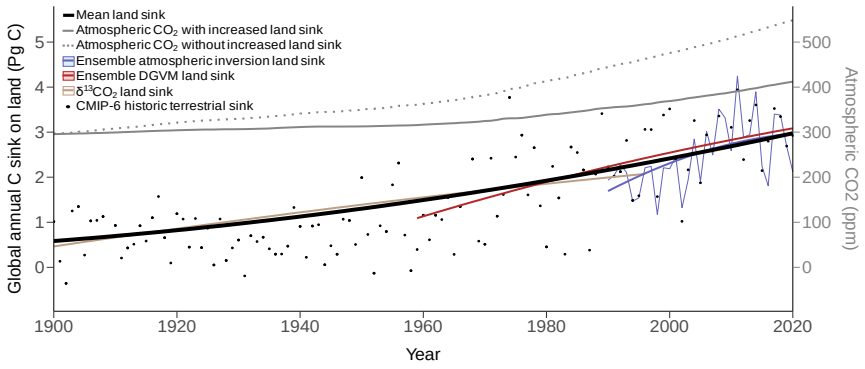
Revealed: more than 90% of rainforest carbon offsets by biggest certifier are worthless, analysis shows

stigation into Verra carbon standard finds most are 'phantom its' and may worsen global heating

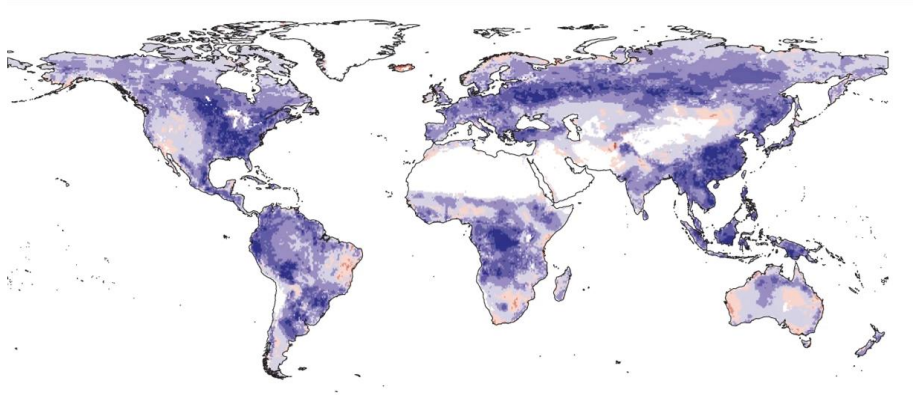
THE GREAT CASH-FOR-CARBON HUSTLE

Offsetting has been hailed as a fix for runaway emissions and climate change—but the market's largest firm sold millions of credits for carbon reductions that weren't real.

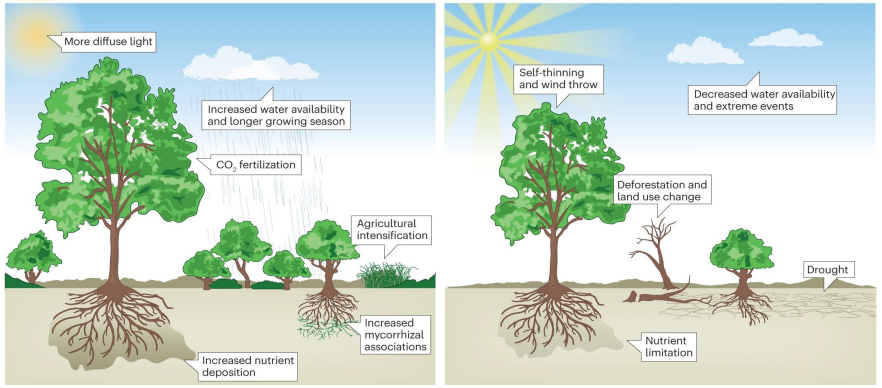
The land carbon sink has increased since 1900



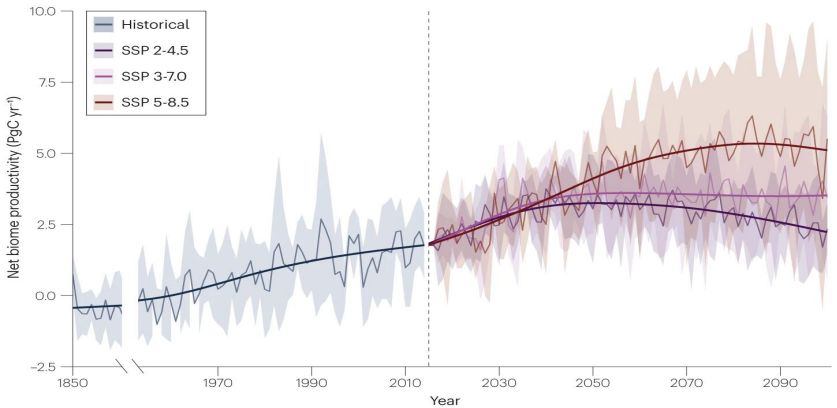
Enhancement arose from longer growing seasons and CO_2 fertilization



Concurrent processes limit and enhance the carbon sink



Although this ecosystem service has large potential, it will saturate

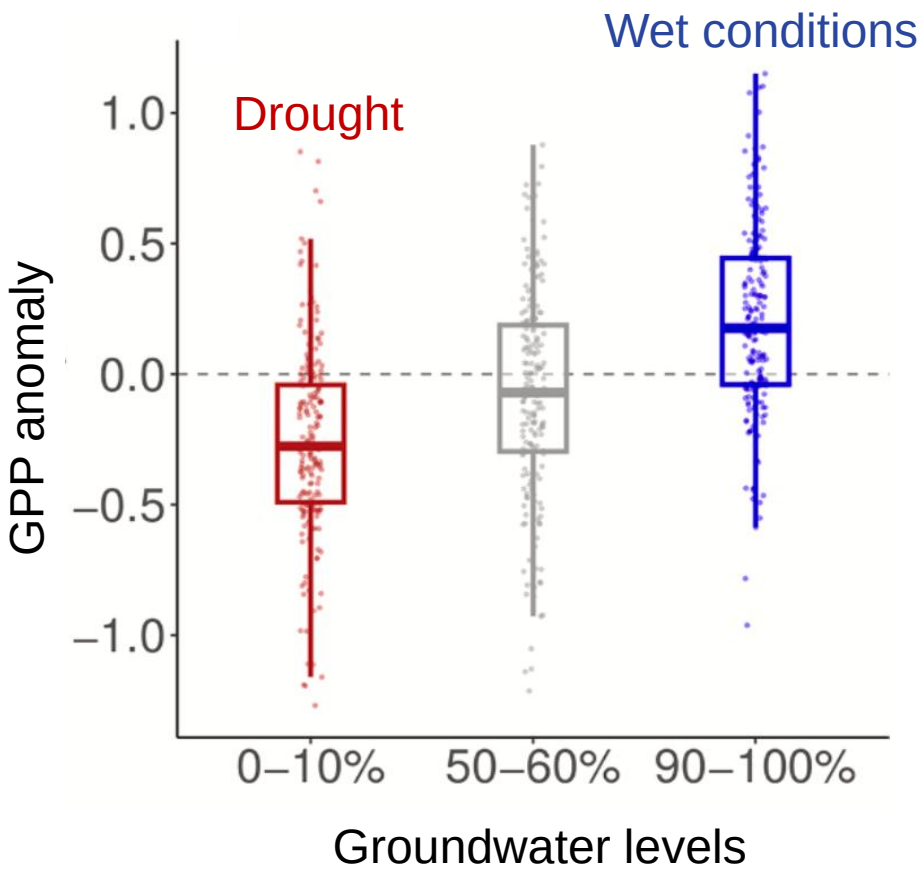


Nature-based climate solutions can help mitigate climate change, but **their longevity is highly uncertain**

Better regulation of carbon markets and **research** to constrain uncertainties is necessary

Centering **Indigenous & marginalized communities** is essential for equitable climate mitigation projects

Ecosystem carbon fixation (GPP) depends on **groundwater drought conditions**



17% increase in annual GPP between drought and wet groundwater conditions

Ruehr et al. Ag For Met, 2023

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Ruehr, S., Keenan, T.F., Williams, C. *et al.* Evidence and attribution of the enhanced land carbon sink. *Nat Rev Earth Environ* **4**, 518–534 (2023).

Roe, S., Streck, C., Beach, R. *et al.* Land-based measures to mitigate climate change: Potential and feasibility by country. *Global Change Biology*, 27, 6025–6058 (2021).

Novick, K., Williams, C., Runkle, B. *et al.* The Science Needed for Robust, Scalable, and Credible Nature-based Climate Solutions in the United States: Full Report (Indiana Univ., 2022).

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